

ibm-granite/**granite-embedding-278m-multilingual**

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Granite-Embedding-278m-multilingual

Model Summary: Granite-Embedding-278M-Multilingual is a 278M parameter model from the Granite Embeddings suite that can be used to generate high quality text embeddings. This model produces embedding vectors of size 768 and is trained using a combination of open source relevance-pair datasets with permissive, enterprise-friendly license, and IBM collected and generated datasets. This model is developed using contrastive finetuning, knowledge distillation and model merging for improved performance.

- **Developers:** Granite Embedding Team, IBM
- **GitHub Repository:** [ibm-granite/granite-embedding-models](#)
- **Website:** [Granite Docs](#)
- **Paper:** [Technical Report](#)
- **Release Date:** December 18th, 2024
- **License:** [Apache 2.0](#)

Supported Languages: English, German, Spanish, French, Japanese, Portuguese, Arabic, Czech, Italian, Korean, Dutch, and Chinese. Users may finetune Granite-Embedding-278M-Multilingual for languages beyond these 12 languages.

Intended use: The model is designed to produce fixed length vector representations for a given text, which can be used for text similarity, retrieval, and search applications.

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Safetensors

Model size 0.3B pa
Tensor type BF16

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Sentence Similarity

Source Sentence

Your sentence here

Sentences to compare

Your sentence here...

Your sentence here...

Add Sentence

Generate

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Maximize

Model tree for ibm-granite/granite-embedding-278m-multilingual

Finetunes 14 models
Quantizations 39 models

Spaces using ibm-granite/granite-embedding-278m-multilingual

ibm-granite/granite-vision-demo

mteb/leaderboard

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Usage with Sentence Transformers: The model is compatible with SentenceTransformer library and is very easy to use:

First, install the sentence transformers library

```
pip install sentence_transformers
```

The model can then be used to encode pairs of text and find the similarity between their representations

```
from sentence_transformers import SentenceTransformer

model_path = "ibm-granite/granite-embedding-278m-multilingual"
# Load the Sentence Transformer model
model = SentenceTransformer(model_path)

input_queries = [
    'Who made the song My achy breaky heart',
    'summit define'
]

input_passages = [
    "Achy Breaky Heart is a country song written by Jeffery Richman and recorded by the band The Judds",
    "Definition of summit for English Language Learners"
]

# encode queries and passages
query_embeddings = model.encode(input_queries)
passage_embeddings = model.encode(input_passages)

# calculate cosine similarity
print(util.cos_sim(query_embeddings, passage_embeddings))
```

Usage with Huggingface Transformers: This is a simple example of how to use the Granite-Embedding-278m-Multilingual model with the Transformers library and PyTorch.

First, install the required libraries

```
pip install transformers torch
```

The model can then be used to encode pairs of text

```
import torch
from transformers import AutoModel, AutoTokenizer

model_path = "ibm-granite/granite-embedding-278m-multilingual"
```

🐙 lablab-ai-amd-developer-hackathon/...

📄 aploetz/granite-embeddings

👤 ipepe/nomic-embeddings

⚡ frozen8569/sahay-ai + 12 Spaces

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Granite Embedding Models
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📊 Evaluation results ⓘ

accuracy ...	test set	self-reported	73.433
f1 on MTE...	test set	self-reported	61.230
f1_weight...	test set	self-reported	78.409
ap on MTE...	test set	self-reported	23.347
ap_weight...	test set	self-reported	23.347
main_scor...	test set	self-reported	73.433
accuracy ...	test set	self-reported	71.806
f1 on MTE...	test set	self-reported	65.647

⌵ Expand 10455 evaluations

```

# Load the model and tokenizer
model = AutoModel.from_pretrained(model_path)
tokenizer = AutoTokenizer.from_pretrained(model_path)
model.eval()

input_queries = [
    ' Who made the song My achy breaky heart',
    'summit define'
]

# tokenize inputs
tokenized_queries = tokenizer(input_queries,
                               padding='max_length',
                               max_length=tokenizer.model_max_length)

# encode queries
with torch.no_grad():
    # Queries
    model_output = model(**tokenized_queries)
    # Perform pooling. granite-embedding-278M
    query_embeddings = model_output[0][:, 0]

# normalize the embeddings
query_embeddings = torch.nn.functional.normalize(query_embeddings, dim=-1)

```

Evaluation: The average performance of the Granite-Embedding-278M-Multilingual on Multilingual Miracl (across 18 languages), Mintaka Retrieval (across 8 languages) and MTEB Retrieval for English (across 15 tasks), German (across 4 tasks), Spanish (across 2 tasks), French (across 5 tasks), Japanese (across 2 tasks), Arabic (1 task), Korean (1 task) and Chinese (across 8 tasks) is reported below.

Model	Parameters (M)	Embedding Dimension	Miracl (18)	Mintaka Retrieval (8)
granite-embedding-278M-multilingual	278	768	58.3	23.2

Model Architecture: Granite-Embedding-278m-Multilingual is based on an encoder-only XLM-RoBERTa like transformer architecture, trained internally at IBM Research.

Model	granite-embedding-30m-english	granite-embedding-125m-english	granite-embedding-107M-multilingual
Embedding size	384	768	384
Number of layers	6	12	6
Number of attention heads	12	12	12
Intermediate size	1536	3072	1536
Activation Function	GeLU	GeLU	GeLU
Vocabulary Size	50265	50265	250002
Max. Sequence Length	512	512	512
# Parameters	30M	125M	107M

Training Data: Overall, the training data consists of four key sources: (1) unsupervised title-body paired data scraped from the web, (2) publicly available paired with permissive, enterprise-friendly license, (3) IBM-internal paired data targetting specific technical domains, and (4) IBM-generated synthetic data. The data is listed below:

Dataset	Num. Pairs
Multilingual MC4	52,823,484
Multilingual Webhose	12,369,322
English Wikipedia	20,745,403
Multilingual Wikimedia	2,911,090
Miracl Corpus (Title-Body)	10,120,398
Stack Exchange Duplicate questions (titles)	304,525
Stack Exchange Duplicate questions (titles)	304,525

Dataset	Num. Pairs
Stack Exchange Duplicate questions (bodies)	250,519
Machine Translations of Stack Exchange Duplicate questions (titles)	187,195
Stack Exchange (Title, Answer) pairs	4,067,139
Stack Exchange (Title, Body) pairs	23,978,013
Stack Exchange (Title, Body) pairs	23,978,013
Machine Translations of Stack Exchange (Title+Body, Answer) pairs	1,827,15
SearchQA	582,261
S2ORC (Title, Abstract)	41,769,185
WikiAnswers Duplicate question pairs	77,427,422
CCNews	614,664
XSum	226,711
SimpleWiki	102,225
Machine Translated Cross Lingual Parallel Corpora	28,376,115
SPECTER citation triplets	684,100
Machine Translations of SPECTER citation triplets	4,104,600
Natural Questions (NQ)	100,231
SQuAD2.0	87,599
HotpotQA	85,000
Fever	109,810
PubMed	20,000,000
Multilingual Miracl Triples	81,409
Multilingual MrTydi Triples	48,715
Sadeeem Question Asnwering	4,037
DBPedia Title-Body Pairs	4,635,922
Synthetic: English Query-Wikipedia Passage	1,879,093
Synthetic: English Fact Verification	9,888
Synthetic: Multilingual Query-Wikipedia Passage	300,266
Synthetic: Multilingual News Summaries	37,489

Dataset	Num. Pairs
IBM Internal Triples	40,290
IBM Internal Title-Body Pairs	1,524,586

Notably, we do not use the popular MS-MARCO retrieval dataset in our training corpus due to its non-commercial license, while other open-source models train on this dataset due to its high quality.

Infrastructure: We train Granite Embedding Models using IBM's computing cluster, Cognitive Compute Cluster, which is outfitted with NVIDIA A100 80gb GPUs. This cluster provides a scalable and efficient infrastructure for training our models over multiple GPUs.

Ethical Considerations and Limitations: The data used to train the base language model was filtered to remove text containing hate, abuse, and profanity. Granite-Embedding-278m-Multilingual is finetuned on 12 languages, and has a context length of 512 tokens (longer texts will be truncated to this size).

Resources

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